Unit - 1

Introduction To CYBERCRIME And Digital forensics

foundation of cybercrime and digital forensics

⇒ **understanding cybercrime:**
- → meaning
- → scope and impact

⇒ **classification of cybercrimes:**
- → Against individuals, organisations.
- → government & society.

⇒ **Cybercrime vs Traditional crime:**
- → Global trends, key differences in nature & evidence and investigation

⇒ **Cyber crime trends and case studies:**
- → Global trends and real world case examples.

Introduction to Digital forensics:
- → Definition, objectives and importance

Types of Digital forensics
- → computer
- → mobile
- → network
- → cloud, database etc

forensics investigation process:
- → overview of investigation stages.

⇒ **Principles of evidence handiling:**
- → Accuracy, intigrity admissibility

→ **chain of custody:**
- → Documentation and tracking of evidence

→ **Locards exchange Principle:**
- → every contact leaves a trace

UNIT 2

* CYBERCRIME TECHNIQUES AND INVESTIGATION PROCESS *

CYBER CRIME INVESTIGATION & ATTACKS LIFE CYCLE

=> Cyber Attack Techniques:
- → Types and mechanisms of attacks.

=> Phishing, malware, Ransomware and Hacking:
- → common attack vectors and impact

=> Social engineering attacks:
- → Baiting, pretexting, tailgating, etc.

=> Cyber fraud methods:
- → online frauds identity theft financial scams.

=> investigation life cycle:
- → Identification
- → collection
- → Preservation
- → Analysis
- → Reporting.

=> Evidence Documentation:
- → Recording, labeling & maintaining logs.

=> open source intelligence (OSINT):
- → Gathering info from public sources

=> Dark web & Deep web Basics:
- → Understanding hidden online resources

UNIT - 3

Digital Evidence and Forensic Acqvistion

ACQUISITION AND PRESERVATION OF DIGITAL EVIDENCE

⇒ Types of Digital Evidence :
→ Documents
→ Emails
→ images, logs
→ multimedia etc.

⇒ Volatile and Non-volatile evedence :
→ Temporary vs permanent data.

⇒ Sources of Evidence :
→ Disk, memory, network logs, mobile devices, cloud etc.

⇒ Hashing Techniqves :
→ MD5, SHA-1, SHA-256 for data integrity.

⇒ Storage medial Analysis :
→ HDD, SSD, USB memory cards, optical media.

⇒ Data Acquisition Techniques :
→ methods to collect digital data

⇒ Live and Dead Acquisition :
→ from running system or from non-operational system.

⇒ Imaging & Duplication - Bit stream copy :
→ creating exact forensic images.

⇒ file systems FAT, NTFS, EXT :
→ Understanding file system structures.

UNIT-4

System, Network and Mobile Forensics

Forensic Analysis Across Systems, Networks and Mobile Devices.

Windows and linux forensics:
→ System artifacts, useractivities, timeline analysis.

Registry and log Analysis:
→ Windows Registry, Event logs, System logs.

Email forensics and Header analysis:
→ Email tracking, header examination, metadata.

Network forensics:
→ network data collection and Analysis.

Packet capture and Traffic Analysis:
→ Tools: Wireshark, tcpdump, etc.

mobile forensics Android and iOS Analysis:
→ Data extraction, app analysis, artifact analysis.

Cloud forensics fundamentals:
→ Cloud data sources, challenges and evidence collection.

malware Analysis Basics:
→ static and dynamic analysis, indicators of malware.

intrution Detection Basics:
→ IDS concepts, alerts and logs.

UNIT - 5

CYBER LAWS, ETHICS AND REPORTING

LEGAL ETHICS & PROFESSIONAL ASPECTS OF DIGITAL FORENSICS.

⇒ Cyber law & it act :-
→ Relevant sections offenses and provisions.

legal Admissibility of digital evidence :-
→ Indian evidence Act, SEC 65B admissibility requirements.

Privacy and Ethical Issues :-
→ Data privacy, ethics in investigation, confidentiality.

Expert witness Role :-
→ Responsibilities in court, testimony & cross-examination

Ⓒ intellectual Property Rights in cyberspace :-
→ copy rights, trademarks, patents and online IPR violations.

⇒ Digital forensics Report writing :-
→ structure, content, clarity and accuracy.

Documentation & presentation
→ charts, graphs, timelines, screenshots and exhibits

Case study & practical investigation scenarios :-
→ Real case analysis and hands-on practice.

CYBER crime & Digital forensics

"from crime to evidence – Towards a safer Digital world"

Unit 1

- → Understanding & classification of cybercrimes
- → Cybercrime vs traditional crime
- → Intro to digital forensics
- → Types of digital forensics
- → forensic investigation process
- → Principles of Evidence Handling

Unit 2

- → Cyber crime attack techniques
- → phishing, malware, ransomware & hacking
- → social engineering attacks
- → cyber fraud methods
- → investigation lifecycle
- → open source intelligence (OSINT)
- → Incident Response Procedures.

Unit 3

- → Types of digital evidence.
- → Data Acquisition Techniques
- → Live & Dead Acquisition
- → imaging & Duplication
- → Hashing techniques
- → Storage media Analysis.
- → file systems (FAT, NTFS, EXT)
- → Data Recovery Techniques

Unit 4

- → window & linux forensics
- → Registry & Log Analysis
- → Email forensics
- → Network forensics.
- → mobile forensics (Andriod & iOS)
- → cloud forensics
- → malware Analysis
- → Intrution Detection Basics.

Unit 5

⇒ Cyber laws And it Act :-

- → legal Admissibility of Digital evidence.
- → Privacy and Ethical issues
- → IPR in cyber space
- → Digital forensic Report writing.
- → case study & practical invastigation.
- → Documentation & Presentation.